H9.

- The assignment is due at Gradescope on 5/23/25.
- A LaTeX template will be provided for each homework. You can either type your homework into this template or scan your handwritten work. If you writing by hand, please fill in the solutions in this template, inserting additional sheets as necessary. This will help facilitate the grading.
- You are permitted to discuss the problems with up to 2 other students in the class (per problem); however, *you must write up your own solutions, in your own words*. Do not submit anything you cannot explain. If you do collaborate with any of the other students on any problem, please list all your collaborators in the appropriate spaces.
- Similarly, please list any other source you have used for each problem, including other textbooks or websites.
- *Show your work.* Answers without justification will be given little credit.
- Algorithms discussed during lecture may be used as black boxes.
- Unfortunately, due to the time constraints of the quarter, there are no resubmittable problems.
- Congrats on making it to the final homework assignment! You've done great :)

Problem 1 (Bottleneck and Lower-Binding Edges) *Let G be a flow network with integer edge capacities. An edge in G is upper-binding if increasing its capacity by 1 also increases the value of the maximum flow in G. Similarly, an edge is lower-binding if decreasing its capacity by 1 also decreases the value of the maximum flow in G.*

(a) *Does every flow network G have at least one upper-binding edge? If so, sketch a proof. If not, give a counterexample.*

(b) *Does every flow network G have at least one lower-binding edge? If so, sketch a proof. If not, give a counterexample.*

(c) *Recall that an edge is saturated under a flow f if $f(e) = c(e)$. Prove or disprove: Given a flow network G, if an edge e is saturated under a max flow f, then e is an upper-binding edge.*

(d) *Given a flow f on a flow network, we define the divergence $div(v)$ at a vertex as:*

$$\sum_{(v,u)\in E} f_{vu} - \sum_{(w,v)\in E} f_{wv}.$$

Sketch a proof that given any subset $S \subseteq V$:

$$\sum_{v\in S} div(v) = \sum_{(u,v)\in E(S,\overline{S})} f_{uv} - \sum_{(u,v)\in E(\overline{S},S)} f_{uv},$$

i.e. the total amount of flow out of S minus the total amount of flow into of S is equal to the total divergence inside of S.

(This is a discrete version of what is known as the divergence theorem in multivariable calculus, though understanding this is not needed to solve this problem).

(e) *Formally prove the following statement: Given a flow network G, if an edge e is an upper-binding edge, then e is saturated under* ***any*** *max flow f.*

Collaborators:

Solution: Your solution here!

Problem 2 (Clique Reductions) *We define a k-Clique as follows:*

Definition 1 (k-Clique) *A k-clique is a set of k vertices $\{v_1, v_2, ..., v_k\}$ in $G = (V, E)$ such that for every $1 \leq i \leq j \leq k$, there is exists an edge $(v_i, v_j) \in E$. In other words, the subgraph formed by the vertices in C form a complete graph.*

Now, consider the following problems:

Definition 2 (Clique) *Given a graph $G = (V, E)$ and a value k, find a k-clique in G.*

Definition 3 (Half-Clique) *Given a graph $G = (V, E)$, find a $(|V|/2)$-clique in G.*

(a) Show that the Half-Clique problem reduces to the k-clique problem.

(b) Show that the k-clique problem reduces to the half-clique problem.

*(c) Using **only** the result from part (b), if half-clique is NP-hard, what guarantees (if any) can we make about the hardness of k-clique?*

Collaborators:

Solution: Your solution here!

Problem 3 (Survey)

1. *What subject(s) did you enjoy the most in the class?*

2. *What subject(s) did you enjoy the least?*

3. *What was your favorite problem or algorithm?*

4. *What problem did you think was the most difficult?*

5. *What would you change in the course?*

6. *Additional comments?*